MyLearna
Plan • Capture • Grow

MATHEMATICS

Name: ____________________

Date: ____________________

Use proportional reasoning

Step 46

Scale up or down, compare fairly, and reason about equivalent relationships.

Today I am learning to use proportional reasoning.

1. Equivalent Ratios and Proportions

Determine whether the ratios form an equivalent proportion.

a) 2 : 3 and 4 : 6 Equivalent / Not equivalent

b) 3 : 5 and 9 : 15 Equivalent / Not equivalent

c) 4 : 7 and 8 : 14 Equivalent / Not equivalent

d) 5 : 6 and 10 : 12 Equivalent / Not equivalent

e) 1 : 2 and 3 : 6 Equivalent / Not equivalent

f) 2 : 5 and 6 : 15 Equivalent / Not equivalent

g) 3 : 4 and 12 : 16 Equivalent / Not equivalent

h) 4 : 9 and 8 : 18 Equivalent / Not equivalent

Remember:

- Equivalent ratios show the same relationship.
- You can find equivalent ratios by multiplying or dividing both parts by the same number.
- Proportion means two ratios are equivalent.

2. Scale Up or Down

Use proportional reasoning to find the missing values.

a) The table shows the number of cups of flour needed to make different batches of muffins.

Batches (x)	1	2	3	4	5
Cups of flour (y)	1.5	3		6	

b) The table shows the number of litres of paint needed to paint different number of rooms.

Rooms (x)	1	2	3	4	5
Litres of paint (y)	2.5		7.5	10	

c) A recipe uses 2 cups of rice to serve 3 people.
How much rice is needed to serve:

i) 6 people? ____________ cups

ii) 9 people? ____________ cups

iii) 12 people? ____________ cups

d) A car travels 180 km with 12 litres of fuel.
How far can it travel with:

i) 6 litres? ____________ km

ii) 18 litres? ____________ km

iii) 24 litres? ____________ km

Remember:

- To scale up or down, multiply or divide both quantities by the same number.
- The relationship between quantities stays the same.
- Use a table to help you see the pattern.

3. Compare Fairly

Use proportional reasoning to compare the quantities fairly.

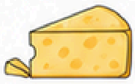
a) A pack of 4 pens costs $3.20.
A pack of 7 pens costs $5.60.
Which pack is better value?
Show your working.

Answer: ____________

b) 3 litres of juice costs $6.00.
5 litres of juice costs $9.50.
Which is better value per litre?
Show your working.

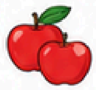
Answer: ____________

c) 500 g of cheese costs $4.50.
800 g of cheese costs $7.20.
Which is better value per 100 g?
Show your working.



Answer: ____________

d) 2 kg of apples costs $6.00.
3.5 kg of apples costs $10.15.
Which is better value per kg?
Show your working.



Answer: ____________

Remember:

- To compare fairly, find the unit rate (cost per 1, per 100 g, per litre, etc.).
- The smaller unit rate means better value.

4. Solve Problems Involving Proportion

Solve the problems using proportional reasoning.

a) A map uses a scale of 1 cm to represent 50 km.
If the distance between two cities on the map is 7.4 cm, what is the actual distance?

Answer: ____________ km

b) A recipe for 4 people uses 3 eggs.
How many eggs are needed for 10 people?

Answer: ____________ eggs

c) A photographer can print 12 photos from a memory card that has 3 GB of storage left. How many photos can he print from a card with 20 GB left? (Assume direct proportion.)

Answer: ____________ photos

d) A school has 240 students. $\frac{3}{5}$ of the students are girls. How many boys are there?

Answer: ____________ boys

Remember:

- Look for the relationship between the quantities.
- Decide whether to scale up or down.
- Keep the units the same on both sides.

5. Real-World Challenge

Use proportional reasoning to solve the challenge.

a) A company can produce 150 notebooks in 3 hours.
How long will it take to produce 500 notebooks at the same rate?

Answer: ____________ hours

b) A family drove 360 km using 24 litres of fuel.
If they continue at the same rate, how many kilometres can they drive with 40 litres of fuel?

Answer: ____________ km

Think about it!

Where do you see proportional reasoning used in everyday life?

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6. Problem Solving

Solve each problem.

a) 5 workers can paint a wall in 6 hours. How long will it take 8 workers to paint the same wall?

Answer: ____________

b) A car travels 270 km in 3 hours. How far will it travel in 7.5 hours at the same speed?

Answer: ____________

c) 4 notebooks cost $3.60. How much will 9 notebooks cost at the same price?

Answer: $ ____________

d) 6 kg of potatoes cost $7.20. How much will 4.5 kg of potatoes cost at the same price per kg?

Answer: $ ____________

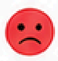
My Learning Check

Circle how confident you feel.

I can do it!

Almost there!

I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands proportional relationships
- ☐ Uses scaling up and down correctly
- ☐ Compares fairly using unit rates
- ☐ Solves real-world problems
- ☐ Shows clear reasoning
- ☐ Works neatly and carefully
- ☐ Needs more practice

Notes: ____________________

Today I learned...

1 thing I did well: ____________________

1 thing I will improve: ____________________